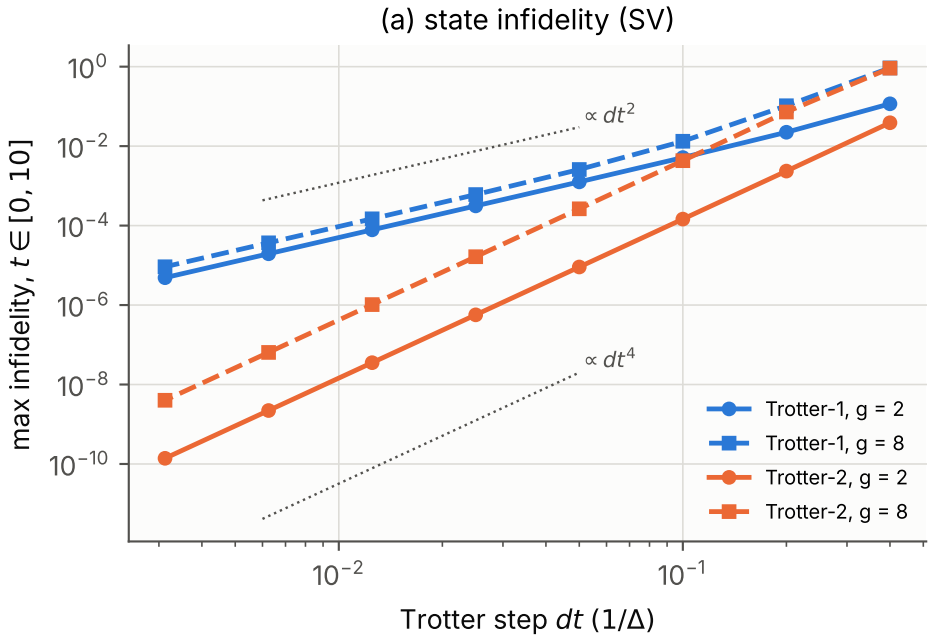
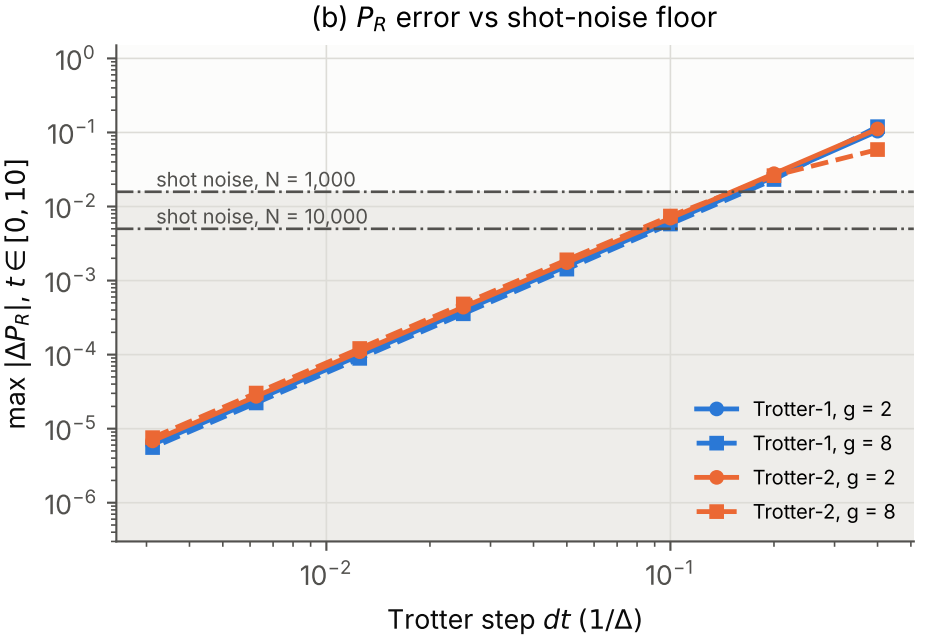

(a) state infidelity (SV)
max infidelity, $t \in [0, 10]$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
$\propto dt^2$
$\propto dt^4$
Trotter-1, g = 2
Trotter-1, g = 8
Trotter-2, g = 2
Trotter-2, g = 8
10^{-2}
10^{-1}
Trotter step dt (1/Δ)


(b) P_R error vs shot-noise floor
max $|\Delta P_R|$, $t \in [0, 10]$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
shot noise, N = 1,000
shot noise, N = 10,000
Trotter-1, g = 2
Trotter-1, g = 8
Trotter-2, g = 2
Trotter-2, g = 8
10^{-2}
10^{-1}
Trotter step dt (1/Δ)